

AI LAB 99

Machine Learning Track — Week 5, Day 4 (Module 10, Part 2)

Quality Assurance Audit Report

Reviewing the Day 4 QA Suite's Coverage, Not Just Its Pass Rate

Source: SQE.ipynb — Final Dataset.csv, kmeans_model.pkl, day3_results.pkl

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Introduction

Day 4 of Week 5 runs a quality-assurance pass intended, per its own header, to test “Dataset, Model, Day 3 Results & Streamlit Outputs” before the project is considered production-ready. This report documents what the suite actually checked, confirms its 6/6 pass result, and then — the more useful part of a QA review — audits what it did not check, several of which are exactly the issues already documented as broken in the Module 8 and Module 10 reports.

Bottom line: the suite's 6 checks all genuinely pass, and “100% pass rate” is an accurate statement about those 6 checks. But the checks are shallow — file-exists, shape, and null checks — not functional or semantic tests, so a 100% pass rate here should not be read as “the pipeline is production-ready.” None of the previously-documented production-blocking bugs (empty strategy matrix, z-score dollar display, non-functional prediction page) would be caught by this suite, and one entire promised test category (Streamlit outputs) was never implemented at all.

File Detection

A recursive search across the project directory located all three required artifacts:

Final Dataset.csv	D:\AI-Lab-99\Week 5\Day 1\Final Dataset.csv — PASS
kmeans_model.pkl	D:\AI-Lab-99\Week 4\Module 6 Customer Segmentation Model Development\ — PASS
day3_results.pkl	D:\AI-Lab-99\Week 3\week4\Module 8 Business Insights and Marketing Recommendations\ — PASS

Note: day3_results.pkl is stored under a folder path that mixes three different week/module labels for what this project otherwise calls Week 5, Day 3. This mirrors the segment-naming inconsistencies already flagged in earlier modules and is worth cleaning up before further files are saved to inconsistent paths.

Dataset Quality Check

Shape	2,038 rows × 42 columns
Duplicate rows	0
Missing values	0
Required columns (9)	Income, Recency, Age, Total_spending, Total_Purchase, Family_size, NumWebPurchases, NumStorePurchases, NumCatalogPurchases — all present

Scale caveat carried forward: the printed sample row shows Income = 0.299651 and similar fractional values across every numeric column — confirming this file is still StandardScaler output, exactly as tracked since Module 7. This QA pass does not flag it, even though it is the root cause the Module 10 report traced the Streamlit dashboard's broken dollar metrics back to.

Model Validation

Model file	Loads successfully as a scikit-learn KMeans object
Expected features	40 (via model.n_features_in_)
Feature names available	Yes, via model.feature_names_in_ — full list of 40 matches the engineered feature set

Minor notebook issue: the model-loading cell (“MODEL VALIDATION”) appears twice in a row with identical code and identical output — harmless, but a sign of copy-paste cell duplication worth cleaning up.

Day 3 Results Validation

day3_results.pkl loaded successfully and was confirmed to contain all 17 expected keys: predicted_cluster, customers_in_segment, activity_level, risk_level, preferred_channel, marketing_message, personalized_offer, campaign_timing, discount_strategy, retention_strategy, primary_product, secondary_product, cross_selling, upselling, segment_summary, marketing_recommendation, and segment_customers.

This check confirms structure, not content. Every key existing in the dictionary is a real pass, but the QA suite stops there — it never inspects what’s inside each value. The Module 8 report already found that some of the underlying tables feeding this exact kind of output (the merged marketing-strategy matrix, in particular) were empty due to a chained-join bug. A dictionary can contain a key whose value is an empty DataFrame and still pass a “key exists” check.

Final QA Summary (As Reported)

Test	Status	Details
Dataset Availability	PASS	Main dataframe loaded successfully
Dataset Not Empty	PASS	Rows: 2,038
Duplicate Check	PASS	Duplicates: 0
Missing Values Check	PASS	Missing values: 0
Cluster Column	PASS	Cluster column present
KMeans Model File	PASS	Found at Week 4 / Module 6 folder

Total Tests: 6 · Passed: 6 · Failed: 0 · Pass Rate: 100%. Results were saved to qa_results.pkl in the Week 5 / Day 4 folder.

Notebook ordering bug (low severity): an earlier cell checks whether results_test_df already exists in memory and prints “No QA results dataframe found yet” — that dataframe is only created in the following cell, so this check is guaranteed to fail on every fresh run. It has no effect on the final result since nothing downstream depends on it, but it's dead/out-of-order code.

What This QA Pass Does Not Cover

The six checks above are necessary but not sufficient for declaring the pipeline production-ready. The table below lists the gaps between what was tested and what a QA pass covering “Dataset, Model, Day 3 Results & Streamlit Outputs” (as promised in the notebook's own header) would need to check:

What a Rigorous QA Pass Would Check	Why It's Missing / What It Would Have Caught
Prediction consistency	No cell re-runs <code>model.predict()</code> on the scaled feature matrix and compares it to the stored Cluster column. As delivered, there is no proof the saved labels actually came from this exact model file.
Streamlit output testing	The Day 4 header explicitly promises to test “Dataset, Model, Day 3 Results & Streamlit Outputs,” but no cell imports, runs, or checks <code>app.py</code> or anything Streamlit-related. This is the one category named in the title that is never actually tested.
Semantic validation of <code>day3_results.pkl</code>	The suite confirms all 17 expected keys exist in the dictionary, but never checks whether their values are meaningful — non-empty, non-identical between segments, etc. The Module 8 report already found several of these same values (campaign response, the merged strategy matrix) to be empty or all-zero; this QA pass would report PASS on that file regardless.
Unit / scale check	Income, Total_spending, and similar columns are still StandardScaler output in this dataset (confirmed by the printed <code>df.head()</code> sample, e.g. Income = 0.299651). The QA suite never flags this, even though it is the exact issue that broke the Module 10 Streamlit dashboard's dollar-formatted metrics.
Segment-name consistency	No check confirms which of the three historical names (“Digital-First Buyers” / “Budget Buyers”) is present in <code>Customer_Segment</code> for this file, despite that being a known, previously-documented inconsistency across modules.

Recommendations

- **Add a prediction-consistency test:** `model.predict(df[feature_names_in_])` compared against `df['Cluster']` with an exact-match assertion, so a QA pass actually proves the saved model produced the saved labels.
- **Add the missing Streamlit test:** at minimum, import `app.py` in a subprocess or use Streamlit's app-testing utilities to confirm it starts without error; ideally, assert that the Dashboard page's income/spending metrics are inverse-transformed before display, so this exact regression can't silently ship again.
- **Add value-level checks to the Day 3 results test:** for each of the 17 keys, assert non-empty and, where applicable, that the two segments' values actually differ — this alone would have caught the empty strategy matrix and the all-zero campaign columns documented in Module 8.
- **Add an explicit scale/units check:** assert that Income's mean is within a business-plausible dollar range (or explicitly assert it is standardized and flag downstream consumers), so scale mismatches like the Module 10 dashboard bug are caught in QA rather than in production.
- **Report pass rate alongside coverage, not instead of it:** a “100% pass, 6 tests” headline reads very differently from “100% pass, 6 of ~15 recommended checks implemented” — the latter gives stakeholders an honest picture of remaining risk.